CYBERSECURITY FUNDAMENTALS

Networking Fundamentals & Linux Fundamentals

October Batch — Practical Assessment

Complete Answer Guide

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Note: Command outputs shown in this guide are illustrative examples. Run commands on your own authorized computer and insert your actual screenshots before submission.

Important note about the network design

The supplied 192.168.10.0/24 network has 256 total addresses and 254 usable host addresses. The requested capacity needs five subnets with block sizes of 128 + 64 + 64 + 32 + 16 = 304 addresses. Therefore, the /24 cannot meet the requirements. The recommended plan uses 192.168.10.0/23 (512 total addresses, 510 usable) subject to approval from the network owner.

Task 1 — Office Network Design Case Study

Q1. Calculate device requirements

Department	Current devices	Calculation (20% extra)	Required capacity
Reception & Administration	20	20 × 1.20 = 24	24
HR & Finance	35	35 × 1.20 = 42	42
Development Team	60	60 × 1.20 = 72	72
SOC & IT Operations	45	45 × 1.20 = 54	54
Server Room	12	12 × 1.20 = 14.4; round up	15

For subnet sizing, choose the smallest power-of-two address block that provides enough usable addresses (network and broadcast addresses are not assignable to ordinary hosts). The server room needs 15 usable addresses, so a /27 is actually required (30 usable); a /28 only provides 14 usable addresses. This means the correct required block sizes are 32 + 64 + 128 + 64 + 32 = 320 addresses, not 304.

Accordingly, use a /23 allocation and the following non-overlapping VLSM plan: /25 for Development, /26 for HR, /26 for SOC, /27 for Reception, and /27 for Servers. Total allocated blocks = 128 + 64 + 64 + 32 + 32 = 320 addresses.

Q2–Q3. Subnetting and IP allocation table

Department	Network / CIDR	Subnet mask	First usable	Last usable	Broadcast	Usable hosts
Development Team	192.168.10.0/25	255.255.255.128	192.168.10.1	192.168.10.126	192.168.10.127	126
HR & Finance	192.168.10.128/26	255.255.255.192	192.168.10.129	192.168.10.190	192.168.10.191	62
SOC & IT Operations	192.168.10.192/26	255.255.255.192	192.168.10.193	192.168.10.254	192.168.10.255	62
Reception & Administration	192.168.11.0/27	255.255.255.224	192.168.11.1	192.168.11.30	192.168.11.31	30
Server Room	192.168.11.32/27	255.255.255.224	192.168.11.33	192.168.11.62	192.168.11.63	30

The table uses only addresses within 192.168.10.0/23 (range 192.168.10.0–192.168.11.255). The subnets do not overlap. Reserve one usable address in each subnet for its router or Layer-3 gateway; assign management IPs to switches and access points from a planned management VLAN/subnet. The five-subnet table is for departmental/server end devices and does not separately define a management VLAN.

Q4. Network capacity analysis

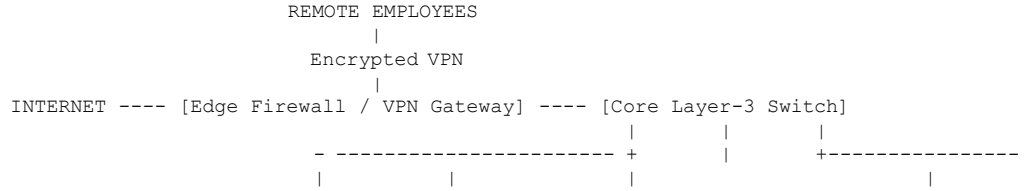
• A /24 has 256 total IPv4 addresses and normally 254 usable host addresses.

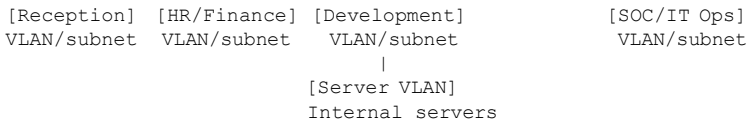
• No. All five subnets with the requested capacity do not fit in one /24. The required subnet blocks total 320 addresses.

• Request a larger allocation, such as 192.168.10.0/23, or use a different approved private range. A /23 has 512 total and 510 usable addresses, enough for this design and some additional room.

• Planning for growth avoids renumbering, service interruption, and hurried network redesign. It also allows addresses for gateways, printers, access points, switches, servers, and other infrastructure.

Q5. Simple network diagram





Deploy a VLAN for each department and a separate server VLAN. Configure routing and access-control lists (ACLs) on the Layer-3 switch or firewall. Remote access should terminate at the VPN gateway, then allow only approved users to reach approved internal resources.

Q6. Security recommendations

• **VLAN and subnet separation:** separates broadcast domains and makes it easier to control department-to-department traffic.

• **Firewall rules / ACLs:** allow only required ports and destinations; deny unnecessary inter-department and employee-to-server traffic.

• **Strong VPN authentication:** use MFA, individual accounts, current VPN software, and least-privilege access; disable unused accounts promptly.

• **Server access restrictions:** permit administration only from authorized IT/SOC systems and log privileged access.

• **Network monitoring:** centralize firewall, VPN, server, and switch logs; alert on unusual traffic and repeated failed logins.

• **Secure Wi-Fi:** use WPA2-AES/WPA3, strong unique credentials or enterprise authentication, updated access points, and a guest network isolated from internal systems.

• **Device hardening and updates:** change default passwords, patch firmware, disable unused services/ports, and back up configurations.

Task 2 — Networking Concepts and Troubleshooting

Q1. Private vs public IP addresses

A private IP is used inside private networks and is not directly routed across the public internet; internet access normally uses NAT. A public IP is globally routable and assigned by an ISP or provider. Private IPv4 ranges are 10.0.0.0/8 and 172.16.0.0/12; the third private range is 192.168.0.0/16.

Q2. What is subnetting?

Subnetting divides a larger IP network into smaller logical networks. It conserves address space, reduces broadcast scope, separates departments, supports security policies, and makes troubleshooting and management easier.

Q3. Default gateway

For 192.168.1.25/24, the host recognizes that a destination outside 192.168.1.0/24 is remote and sends the packet to its configured gateway, 192.168.1.1. The gateway routes it toward the destination network. The gateway must be reachable on the local subnet.

Q4. DNS problem

Possible causes include a wrong DNS server, DNS service outage, bad cache, incorrect domain record, or firewall blocking DNS (usually UDP/TCP port 53). Investigate with nslookup or dig, compare results from another DNS server where authorized, check network configuration, flush the local DNS cache if appropriate, and verify whether other domain names resolve.

Q5. DHCP

DHCP automatically provides IP address, subnet mask, default gateway, DNS servers, and lease duration. If DHCP fails, a host may receive no address or an APIPA address such as 169.254.x.x on Windows, and may lack connectivity. Check link/Wi-Fi, DHCP server availability, scope exhaustion, VLAN/relay configuration, and client lease status.

Q6. TCP vs UDP

TCP is connection-oriented and provides ordered, reliable delivery with retransmission; examples include HTTPS and SSH. UDP is connectionless with lower overhead and no built-in delivery guarantee; examples include DNS queries and real-time voice/video. Application behavior can vary by protocol version.

Q7. VPN

A VPN creates an encrypted tunnel between the remote employee and company network, protecting traffic over untrusted networks. Access should use MFA and least privilege. A poorly configured VPN can expose internal services too broadly or allow compromised remote devices into sensitive networks.

Q8. Local network works, internet fails

Check in order: (1) physical/Wi-Fi link and IP configuration; (2) ping the default gateway; (3) inspect routing table; (4) ping a known external IP to separate routing from DNS; (5) test DNS with nslookup; (6) inspect firewall/proxy/VPN settings; (7) use traceroute/tracert to locate where the path stops. Some hosts block ping, so corroborate with other tests.

Task 3 — Practical Networking Commands

Run the commands on your own computer or authorized lab machine. The output depends on your actual system, so the examples below are templates rather than claimed results.

Activity 1: Identify network configuration

Purpose	Linux	Windows
IP, mask, interface	ip addr	ipconfig /all
Gateway and routes	ip route	route print
DNS settings	resolvectl status (if available); cat /etc/resolv.co	nipconfig /all
Hostname	hostname	hostname

Activity 2: Connectivity tests

```
# Linux
ip addr
ip route
ping -c 4 <your-default-gateway>
ping -c 4 1.1.1.1
nslookup example.com
traceroute example.com       # or tracepath example.com

# Windows (Command Prompt)
ipconfig /all
ping <your-default-gateway>
ping 1.1.1.1
nslookup example.com
tracert example.com
```

Activity 3: Interpret results

• Successful ping: ICMP echo replies returned, showing IP reachability and round-trip times at that moment.

• A failed ping does not always mean unreachable: a firewall or host may block ICMP while the service itself works.

• DNS lookup translates a domain name to records such as IP addresses and helps identify name-resolution failures.

• Traceroute shows the sequence of responding routers/hops and approximate delay; some hops may not respond.

• A device can have an IP but no internet because the gateway, route, DNS, firewall, ISP, Wi-Fi uplink, or captive portal is failing.

Evidence to add: capture your own terminal/Command Prompt screenshots for the commands you ran. Do not submit these illustrative commands as if they were your actual screenshots.

Task 4 — Linux Filesystem Navigation and Commands

Activity 1: Identify your environment

```
pwd
whoami
hostname
uname -r
date
```

These display the current working directory, logged-in user, hostname, kernel release, and current date/time respectively.

Activity 2: Important directories

Directory	Purpose	Example
/	Root of the filesystem hierarchy; all paths begin here.	/home, /etc
/home	Personal directories for regular users.	/home/student/Documents
/etc	System-wide configuration files.	/etc/hosts, /etc/ssh/
/var	Frequently changing data such as logs and caches.	/var/log/
/tmp	Temporary files; cleanup policy depends on system.	Temporary application files
/usr	Most user-space programs, libraries, and shared data.	/usr/bin/
/bin	Essential user commands; on many modern systems this is a symlink to /usr/bin.	/bin/ls
/root	Home directory of the root administrator.	/root/.bashrc

Activity 3: Create the directory structure

```
mkdir -p ~/October_Assessment/{Networking,Linux,Reports}
touch ~/October_Assessment/Networking/subnetting.txt
touch ~/October_Assessment/Networking/network_notes.txt
touch ~/October_Assessment/Linux/commands.txt
touch ~/October_Assessment/Linux/filesystem.txt
touch ~/October_Assessment/Reports/assessment_summary.txt

find ~/October_Assessment -maxdepth 3 -print
ls -la ~/October_Assessment
realpath ~/October_Assessment
cd ~/October_Assessment/Linux       # absolute or home-based path
cd ../Networking                   # relative path from Linux
cd ~/October_Assessment/Reports
```

The diagram in the supplied assessment shows network_notes.txt under Networking. The commands above create it at that location.

Activity 4: File operations

```
cd ~/October_Assessment
printf 'Linux commands help manage files and inspect the system.\n' > Linux/commands.txt
cat Linux/commands.txt
printf 'Commands should be used carefully and with appropriate permissions.\n' >> Linux/commands.txt
cp Linux/commands.txt Linux/commands.txt.bak
mv Linux/filesystem.txt Linux/linux_filesystem.txt
cp Networking/subnetting.txt Reports/subnetting.txt
touch temp_remove_me.txt
rm temp_remove_me.txt
```

Command explanations: `printf ... >` creates/overwrites a file; `>>` appends; `cat` displays content; `cp` copies; `mv` renames/moves; `rm` removes a file. Use rm carefully and only on the temporary file you created.

Absolute path: starts at /, for example /home/student/October_Assessment. **Relative path:** is interpreted from the current directory, for example ../Networking.

Task 5 — Linux Permissions and File Management

Activity 1: Understand permissions

```
cd ~/October_Assessment
touch network_report.txt
ls -l network_report.txt
```

• r = read, w = write, x = execute. For a regular file, execute means it can be run as a program/script; for a directory, x means traversal/search permission.

• File permissions control reading, modifying, and executing a file. Directory permissions control listing names (r), creating/removing/renaming entries (w, usually combined with x), and traversing/accessing entries (x).

• Owner is the user who owns the file; group is the associated group; others means all other users.

• rw-r--r-- means owner can read/write; group can read; others can read. It is commonly represented as 644.

• A directory has an execute bit for traversal; for example, drwxr-xr-x means the owner can list/create/remove entries (with suitable permissions), while group and others can list and traverse but not modify entries.

Activity 2: Modify permissions

```
chmod 640 network_report.txt
ls -l network_report.txt
chmod 600 network_report.txt
ls -l network_report.txt
```

640 = rw-r-----: owner can read/write (6), group can read (4), others have no access (0). 600 = rw-------- only the owner has read/write permission; group and others have no permissions. Actual access can also be affected by ACLs, parent-directory permissions, and privileges.

Activity 3: Ownership

```
ls -l network_report.txt
stat network_report.txt
# Example only; use a real account/group you are authorized to assign:
sudo chown <username>:<groupname> network_report.txt
```

• `ls -l` and `stat` show owner and group information.

• `chown` changes ownership. It often requires administrator privileges because changing ownership affects system security and accountability.

• Sensitive reports should be restricted to reduce unauthorized disclosure, alteration, or misuse.

Do not run the example chown command literally with angle-bracket placeholders. Do not change system-file ownership; only use files you created and accounts/groups you are authorized to manage.

Activity 4: Search and inspect files

```
find ~/October_Assessment -type f -name '*.txt'
grep -n 'network' ~/October_Assessment/Linux/commands.txt
head -n 5 ~/October_Assessment/Linux/commands.txt
tail -n 5 ~/October_Assessment/Linux/commands.txt
wc ~/October_Assessment/Linux/commands.txt
file ~/October_Assessment/Linux/commands.txt
ls -l ~/October_Assessment/network_report.txt
```

Task 5 questions

• **chmod vs chown:** chmod changes permission bits; chown changes the file's owner and/or group.

• **Why 777 is poor for sensitive files:** it gives everyone read, write, and execute permissions, allowing unauthorized changes or execution.

• **> vs >>:** > redirects output and replaces existing file content; >> appends output to the end.

• **Standard output vs standard error:** stdout (file descriptor 1) is normal command output; stderr (descriptor 2) is diagnostic/error output. They can be redirected separately.

• **Least privilege:** give users and processes only the access needed for their work, limiting accidental damage and the impact of compromised accounts.

Task 6 — Linux Mini Challenge: Investigate an IT Environment

Step 1: Create the lab and synthetic log

```
cd ~/October_Assessment
mkdir -p Linux_Investigation/{logs,reports,backup}
touch Linux_Investigation/notes.txt
cat > Linux_Investigation/logs/application.log <<'EOF'
2026-10-16 09:10:12 INFO Service started
2026-10-16 09:11:03 INFO Configuration loaded
2026-10-16 09:12:45 ERROR Failed to connect to database
2026-10-16 09:13:10 WARNING Retrying connection
2026-10-16 09:14:02 ERROR Database connection timeout
2026-10-16 09:15:30 INFO Service stopped
EOF

printf 'This report is used to test Linux file permissions.\n' > Linux_Investigation/reports/network_report.txt
```

Step 2: Investigate files

```
cat Linux_Investigation/logs/application.log
grep 'ERROR' Linux_Investigation/logs/application.log
grep -c 'ERROR' Linux_Investigation/logs/application.log
tail -n 3 Linux_Investigation/logs/application.log
find Linux_Investigation -type f
cp Linux_Investigation/logs/application.log Linux_Investigation/backup/application.log
ls -l Linux_Investigation/backup
```

Expected result for the provided synthetic log: there are **2** ERROR entries. The messages are “Failed to connect to database” and “Database connection timeout.”

Step 3: Change and verify permissions

```
ls -l Linux_Investigation/reports/network_report.txt
chmod 600 Linux_Investigation/reports/network_report.txt
ls -l Linux_Investigation/reports/network_report.txt
stat Linux_Investigation/reports/network_report.txt
```

With mode 600, the owner can read and modify the report; group members and other users receive no permission through the standard mode bits. Parent directory permissions and special ACLs may also affect access.

Step 4: Create investigation summary

```
cat > Linux_Investigation/investigation_summary.txt <<'EOF'
Investigation summary
Error entries found: 2
Error messages:
- Failed to connect to database
- Database connection timeout
Backup file: Linux_Investigation/backup/application.log
Report permissions: 600 (rw------ )
Commands used: cat to display the log, grep to search errors, grep -c to count them,
tail to show the last lines, find to list files, cp to create a backup,
chmod to restrict report permissions, and ls/stat to verify permissions.
EOF

cat Linux_Investigation/investigation_summary.txt
find Linux_Investigation -maxdepth 3 -print
```

Task 6 questions

• **1. Search errors:** grep 'ERROR' application.log.

• **2. Count matching lines:** grep -c 'ERROR' application.log.

• **3. Why keep a log backup?** It preserves evidence for investigation and recovery if the original is changed, rotated, or removed.

• **4. Permission 600:** owner read/write; no group or other permissions.

• **5. File exists but cannot open:** inspect ls -l, stat, owner/group, mode bits, ACLs (getfacl if available), parent-directory execute permissions, and any security policy or mount restrictions.

• **6. Missing vs inaccessible:** use ls or test -e to check existence, and inspect the exact error. “No such file or directory” suggests a missing path component; “Permission denied” suggests an access restriction. Lack of permission to a parent directory can also hide a file from listing.

Conclusion

This assessment demonstrates how to plan a scalable segmented office network, explain core networking services, test connectivity, navigate the Linux filesystem, manage files and permissions, and investigate synthetic application logs. Before submission, run the commands on your own authorized lab machine, add genuine screenshots and actual outputs, and review every command and explanation so you can explain your decisions.